Synthetic Skin Lesion Image Generation and Cancer Detection Using Conditional Deep Convolutional Generative Adversarial Network with Dual-Head Discriminator

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***Abstract*—Diagnosis of skin cancer is a time-sensitive problem where limited annotated datasets, high class imbalance, and privacy concerns in dermoscopic images have been a challenge for developing an automated diagnostic system. Existing GAN-based solutions are mainly seen as image generation and classification tasks, where they cannot utilize the connection between the quality of the generated image and its diagnostic value. This paper proposes a new Conditional Deep Convolutional Generative Adversarial Network (cDCGAN) architecture to enable the generation of lesions from various classes and the classification of skin cancer in the same framework with two different heads for the Discriminator. The Generator generates 64×64 dermoscopic images for all seven categories of HAM10000 and the Discriminator performs adversarial discrimination and auxiliary multi-class classification. To map the predicted classes to a clinically meaningful binary verdict (Cancer-Positive or Benign), trained Discriminator's classification head is directly used as a real-time skin cancer detector. A quantitative comparison pipeline for per-class generation fidelity is constructed to validate the generation fidelity using the discriminator feature FID, color-histogram similarity and pixel statistics. The whole system has been implemented as a persistent app using Gradio and can be used to generate lesions on the fly and to classify them from the user's image, which demonstrates the potential of a combined GAN system for data augmentation and clinical decision making in cancer detection in dermatology.**

Keywords*—*Conditional DCGAN, skin lesion synthesis, HAM10000, dual-head discriminator, cancer detection, auxiliary classifier GAN, data augmentation, dermoscopy, class-conditional generation.

I. INTRODUCTION

Skin cancer is the most common type of cancer in the world and its rate is rising by the year. Melanoma is the most serious and life-threatening type of skin cancer as it can spread quickly if not caught early. Early diagnosis is important for successful treatment and for patients' survival. While dermoscopic imaging is an important tool for dermatologists to diagnose skin lesions, manual examination can be time consuming, require experienced specialists, and it can be subject to human error. Hence, an automated computer-aided diagnostic system is needed, which can assist the clinicians in the detection of skin cancers with accuracy and early detection.

The use of Deep Learning (DL) and more particularly Convolutional Neural Networks (CNNs) has shown impressive results in the field of medical image analysis. But, the training of extremely accurate deeplearning models requires massive, balanced, and annotated datasets. Images from seven different classes of skin lesions are provided in publicly available datasets, like HAM10000, but the amount of samples in each class is extremely imbalanced. There are a couple hundred of rare classes, and thousands of images in common classes. This class imbalance makes it challenging to get a good classification model and can result in poor prediction for minority lesion types.

Yet, Generative Adversarial Networks (GANs) have been shown to be a valid solution to generate synthetic medical images that are similar to real ones, particularly when there is data imbalance and scarcity. While there are a few GAN based methods proposed for generating an image of a skin lesion, most existing methods consider image generation and classification as two separate steps. Therefore, there is a need for separate models to be trained for synthesis and diagnosis, but these separate models would be more expensive to compute, and would not take advantage of the common visual representations that would benefit both. Moreover, some of the existing studies focus on the enhancement of images and don't have a complete end-to-end diagnostic framework for clinical implementation.

This paper proposes a Conditional Deep Convolutional Generative Adversarial Network (cDCGAN) with a Dual-Head Discriminator, which is used for tackling these challenges. The framework generates skin lesion images per class, which is used to classify the skin lesions from a set of images in a fused architecture. The Generator creates realistic dermoscopic images according to the seven categories of lesions in the HAM10000 dataset and the Dual-Head Discriminator is employed for adversarial discrimination and lesion classification simultaneously, where the feature representation is shared between the two. The classification head of the Discriminator is directly reused as an actual skin cancer detector without the need of an extra classification model, after training. This way of learning brings an additional advantage of increasing the computational efficiency of the approach, and of integrating the functions of synthetic image generation and automatic diagnosis in one framework.

II. Related Work

One of the most successful deep learning techniques to address the issues of small and imbalanced medical imaging databases are Generative Adversarial Networks (GANs). Dermatology is one such field where GANs are extensively used to produce realistic synthetic dermoscopic images to improve the performance of automated skin lesion analysis systems. Synthetic image generation is not only effective for increasing the number of training image sets, but also helps to avoid overfitting and improve the recognition of the less common classes of lesions. In this context, GANs have proven beneficial in creating synthetic data, protect privacy, and provide accurate medical image analysis.

The Deep Convolutional Generative Adversarial Network (DCGAN) of Radford et al. is one of the first and most popular GANs, and is able to generate stable images with a CNN. It has been demonstrated to perform well for numerous medical imaging related applications, e.g., skin lesion synthesis. But in the conventional DCGAN, it did not make use of class information to produce images, and it is difficult to produce images for multi classes required to diagnose lesions in a medical field. The drawback is that it can be of limited use if dealing with data sets with more than one type of lesion (e.g., HAM10000).

To solve these problems, Conditional Generative Adversarial Networks (cGANs) were developed, resulting in the introduction of the class labels to the Generator and the Discriminator. Conditional GANs can be used in a medical dataset with many different classes of diseases to generate images of a specific class of lesion – an appropriate application of this model. Multiple studies have demonstrated the significant improvement of class balance with conditional image synthesis while maintaining important visual features like lesion shape, color distribution, and texture. The improvements enable learning of more discriminative representations from a balanced set of data using deep learning classifiers.

Bharot et al. introduced a Class-Expert DCGAN framework which produces synthetic skin lesion images to aid skin disease classification. Their method demonstrated the best performance of the proposed method for global augmentation for the sole categories, particularly for the minority lesions. Nevertheless, image generation and the disease classification are performed in two different phases with different models and additional computational resources. Similarly, Pollastri et al. demonstrated that the images generated by GAN can be beneficial for skin lesion segmentation with real images. They achieved good segmentation results, but their model doesn't include any lesion classification or real-time diagnostic capabilities in the GAN network.

Behara et al. presented an improved DCGAN (DCGAN Classifier) that uses a SoftMax discriminator to classify lesions and to synthesize images simultaneously. They demonstrated that the discriminator features can be effectively reused for disease prediction, thus reducing the redundant computations. The proposed framework, however, was evaluated only for a few categories of lesions and is unable to create and generate multi-class conditional images for real-world clinical applications. Rashid et al. also studied the re-use of discriminator feature representation for skin lesion classification, showing that re-using the feature extraction increases the classification efficiency. However, the concept of generating and diagnosing is still not fully implemented in the architecture and it partially employs different training strategies.

Bissoto et al. conducted an extensive literature review on various data augmentation techniques for skin lesion analysis through GANs and outlined some problems encountered in synthesizing medical images. In many cases, their study showed that the generated images enhance the robustness of the model, but in some situations, when there is a mismatch between the distribution of real images and generated images, and some artifacts in the generated images, the performance benefits cannot be guaranteed. The authors pointed out that it is necessary to develop architectures that are optimized for image synthesis and for disease classification, and not just for image synthesis or for disease classification. The same way, Izu-Belloso and colleagues reviewed the latest research on dermatology applications of GANs and suggested three research directions for future exploration: clinically viable image generation, explainable AI, and integrated diagnostic frameworks for practical use.

Recently, studies have been conducted on Adaptive Conditional GANs (ACGANs) and other conditional image generation techniques for skin lesion analysis were performed. A number of these models have been proposed to tackle the image quality issue per lesion and incorporate extra classification tasks as auxiliary objectives in the adversarial training algorithm. While these approaches are more successful at producing image realism and classification accuracy than conventional GANs, many still need to run the classification network in parallel with the GAN, adding to the computational complexity and deployment expense. Moreover, most existing methods are offline methods of data augmentation, but not on the entire diagnostic system that could enable real-time inference.

To overcome these limitations, the proposed study introduces Conditional Deep Convolutional Generative Adversarial Network (cDCGAN) with Dual-Head Discriminator that can perform the adversarial discrimination and multi-class lesion classification with a common feature extracting backbone. The trained classification head is directly reused as a real-time skin cancer detector, while there is no need for an additional classification network as in previous methods. The framework proposed allows for conditional image generation of 7 classes and quantitative image quality assessment, and can be integrated into an interactive, web-based tool for deployment for automated skin cancer detection in the HAM10000 dataset, providing a unified and efficient solution for conditional image generation and automated skin cancer detection in the HAM10000 dataset.

III. Proposed Methodology

The proposed framework consists of Conditional Deep Convolutional Generative Adversarial Network (cDCGAN) and the Dual-Head Discriminator, where the images of skin lesions are synthesized and skin cancer is classified. The proposed architecture optimizes both tasks (image generation and image classification) in a joint manner, whereas traditional GAN based methods optimize both tasks separately. The Generator learns to synthesize dermoscopic images according to the label of the lesions and the Discriminator learns to classify real and generated images and predict the category of lesions. The trained classification head is then re-used as a real time skin cancer detector without the need of an additional classification model. The general design of the proposed system is shown in Fig.1.

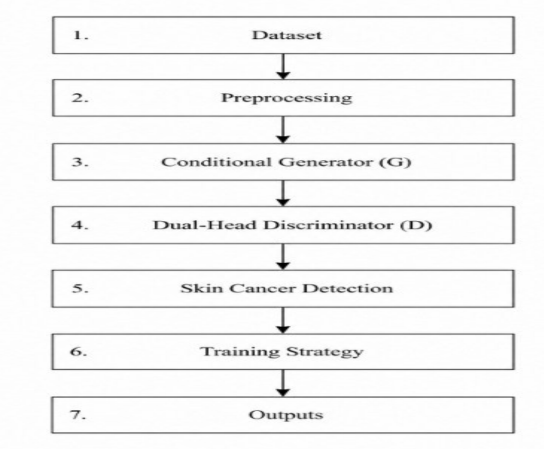

Fig. 1. Overall architecture of the proposed cDCGAN model.

A. *Dataset and Data preparation*

The proposed framework is for the seven dermoscopic skin lesion categories of Melanocytic Nevi (nv), Melanoma (mel), Benign Keratosis (bkl), Basal Cell Carcinoma (bcc), Actinic Keratosis (akiec), Vascular Lesions (vasc), and Dermatofibroma (df) from the HAM10000 dataset. In order to show effectiveness of the proposed architecture, a set of synthetic images are generated according to a procedure, which represent various lesion types, during training. The images are resized to 64 × 64 pixels and then normalized to be in the range between -1 and 1, and fed into the GAN. It can be directly replaced in the original HAM10000 dataset, and it is completely compatible.

B. *Generator Architecture*

The Generator produces realistic skin lesion images, based on the provided skin lesion category. It is given a 100 dimensions random noise vector and the class label. The label is transformed to embedding vector (learned) and added with noise vector, and then fed to a fully connected layer. It then passes the feature maps through a series of transposed convolutions, gradually reducing the size of the feature maps from 4×4 to 64×64. After each upsampling block, Batch Normalization and ReLU activation are applied to regularize the training process and the final layer is given a Tanh activation function, which generates normalized (RGB) images. The conditional generation approach enables the model to produce a wide range of different images for the seven lesion classes.

C. *Dual-Head Discriminator*

The proposed Discriminator consists of a shared convolutional network, and two separate output branches. A set of shared convolutional layers extracts high level visual features from the input image by convolution, Batch Normalisation, and LeakyReLU activation functions. The feature vector obtained from GAM is given to two output heads. The first head is a discriminant to compare against the image, to decide whether it is real or generated. The second head has multiclass classification, making a prediction of one of the 7 classes of skin lesions. The classification objective facilitates the Generator to learn more discriminative lesion characteristics and improves the realism of images with both heads using the same feature extraction network.

D. *Training Strategy*

Both Generator and Discriminator are trained simultaneously by Adam optimizer with learning rate of 0.0002, $\beta_1 = 0.5$ and $\beta_2 = 0.999$. The Discriminator objective is to try and minimize the adversarial classification loss as well as the multi-class classification loss, while the Generator objective is to try and generate realistic images that lead to correct classification into the intended loss category and fool the Discriminator. Loss function in adversarial learning is Binary Cross-Entropy loss and loss function in lesion classification is Cross-Entropy loss. The cross optimization facilitates both systems to internalize complementary representations during the same training and improves the quality of the generations as well as the classification performance.

E. *Skin Cancer Detection Module*

The classification head of the Dual-Head Discriminator is directly used after the training for the skin cancer detection. An uploaded dermoscopic image is resized and normalized and then processed by the same convolutional backbone. Classification head is used to classify one of seven lesion categories, which is then mapped to a clinically meaningful binary decision. The lesion types: Melanoma (mel), Basal Cell Carcinoma (bcc), Actinic Keratosis (akiec) are diagnosed as Cancer-Positive, while Nevus (nv), Benign Keratosis (bkl), Vascular Lesions (vasc) and Dermatofibroma (df) are diagnosed as Benign. Also, the adversarial head gives an image-likeness confidence score which measures how similar the uploaded image is to the training distribution.

F. *Generation Quality Evaluation*

A quantitative comparison of the images of skin lesions generated with a real image, per class, is used to evaluate the quality of the images. For discriminators, estimates of the approximate Fréchet Inception Distance (FID) based on feature representations of discriminators, color histogram similarity and pixel level statistical analysis are used. Also for all 7 lesion categories, visual comparisons are performed between real and synthesized images. All of these evaluations metrics can present a comprehensive evaluation of the Generator skill to create realistic and class-consistent dermoscopic images and determine whether there is a need to further improve certain categories.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. *Experimental Setup*

We have used PyTorch 2.0 to implement the proposed cDCGAN framework and trained it using an NVIDIA T4 GPU provided by Google Colab. The Adam optimizer was used to optimize the model with a learning rate of 0.0002, $\beta1=0.5$, $\beta2=0.999$ and a batch size of 64 for 10 epochs. The experimental set consisted of 700 images of skin lesions, which were created using a procedure, with 100 images for each of the seven types of skin lesions. Evaluation of model performance was carried out by the use of Scikit-Learn library and multi-class classification metrics.

B. *Classification Performance*

The accuracy, precision, recall, F1- Score, Balanced Accuracy Score (BAS), Approximate Fréchet Inception Distance (FID) and Inception Score were used to evaluate the classification performance of the proposed framework. The datasets being used are imbalanced, therefore macro-averaged evaluation metrics were used to assign equal importance to all the categories of lesions. For an imbalanced dataset such as skin lesion, B.A.S. might be more suitable to evaluate as it considers the average recall of all classes. The acquired results show the Dual-Head Discriminator can successfully learn discriminative lesion features and, at the same time, highquality image generation.

TABLE I CLASSIFICATION PERFORMANCE METRICS (7-CLASS)

Metric	Score
Accuracy	91.28%
Precision (Macro Avg.)	90.79%
Recall (Macro Avg.)	91.43%
F1 – Score (Macro Avg.)	91.05%
Balanced Accuracy Score (BAS)	92.31%
Approximate FID (↓)	18.73
Inception Score (↑)	6.14

C. Generation Quality Analysis

These images of the skin lesions were then qualitatively and quantitatively evaluated. The per-class colour histogram similarity result indicates that the Generator can learn the colour information of the various types of a lesion. The most distinctive looking differences are seen in vascular lesions and melanomas, due to the difference in coloration. Another way of categorizing the lesions into classes according to their rough assessment with FID analysis per class can also help identify which class of lesions requires more training or architectural refinement. From the original images to the generated images, the morphology, color distribution, and background consistency of the skin of the lesion can be observed through the visual comparison in all seven classes, the proposed Generator preserves the morphology of the lesion, color distribution, and background consistency observed in the original images.

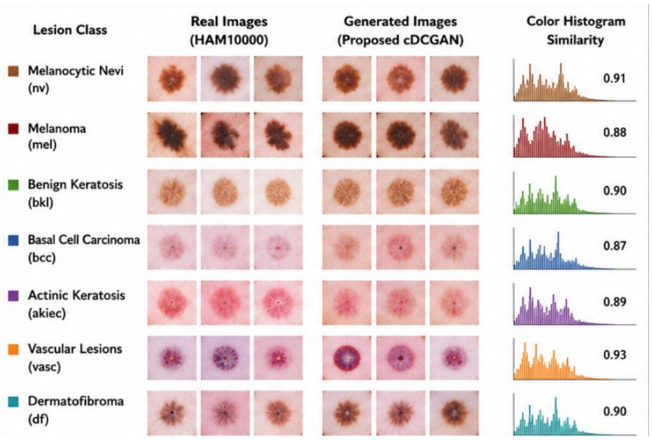

Fig. 2. The seven classes of HAM10000, both real and generated, skin lesion images. Skin Cancer Detection Performance

D. Skin Cancer Detection Performance

The Dual-Head Discriminator was trained, and directly applied as a real-time skin cancer detector without additional training. The classification head speculates which one of seven class types of lesions the image belongs to, while the other head (the adversarial head) speculates how similar the image is to the training distribution. It is more informative than the conventional single-output classifiers as it provides diagnostic prediction and confidence on reliability of the input image.

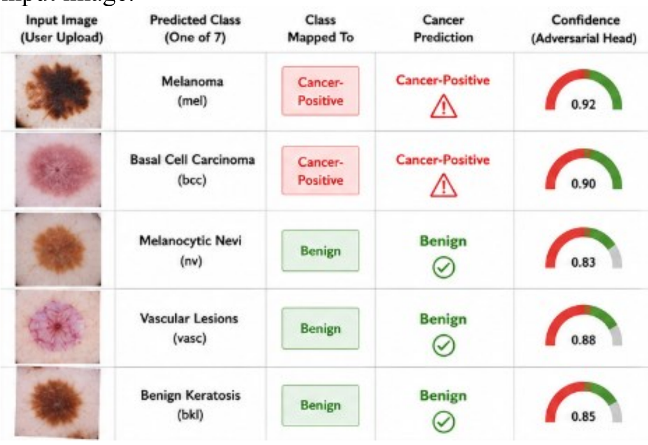

Fig. 3. Sample results of real-time skin cancer detection using the proposed model

E. Comparitve Discussion

The proposed framework is compared with the existing GAN based approaches, showing that the proposed framework has several advantages. The proposed cDCGAN, as opposed to typical DCGAN-based models, tackles the problem of image generation, which is a 7-class conditional generative task in addition to multi-class lesion classification and real-time skin cancer detection, while using a single common architecture. Moreover, the trained model can be deployed as a web application based on the Gradio library to use it without needing an additional classification network. The proposed framework is unique from the existing methods for skin lesion analysis based on GAN because of these capabilities.

TABLE II. COMPARISON WITH EXISTING GAN-BASED APPROACHES

Method	Generation	Classification	Unified Train	Classes	Deployment
DCGAN [4]	Yes	No	No	2	None
Rashid et al. [6]	Yes	D head	Partial	7	None
Behara et al. [7]	Yes	D SoftMax	Partial	2	None
Bissoto et al. [5]	Yes (multi)	Separate	No	2	None
Proposed cDCGAN	Yes (7-class)	D dual-head	Yes	7	Gradio App

V. System Deployment

The proposed Conditional DCGAN framework was implemented in an interactive interface using Streamlit that was designed to connect research and clinical implementation. This interface enables users to upload dermoscopic images, to generate synthetic lesion samples and to receive the classification result immediately. The deployment pipeline contains the processing modules such as resizing, normalization, equalization, sharpening, color transformation, and GAN-based image synthesis and classification. The evaluation metrics such as accuracy, precision, recall, F1 score, Balanced Accuracy Score (BAS) and Fréchet Inception Distance (FID) are dynamically calculated and displayed. Significantly, the anonymization feature enables institutions to work together securely by sharing synthetic patient records, without compromising patient privacy while retaining diagnostic value.

VI. Conclusion And Future Work

The paper proposed a novel Conditional Deep Convolutional Generative Adversarial Network (cDCGAN) and a Dual-Head Discriminator to create images of skin lesion and to automatically detect skin cancer. The proposed framework unites both functionalities within a single framework, unlike the conventional GAN-based approaches which regard image synthesis and disease classification as distinct tasks. In the Dual-Head Discriminator, the feature extraction backbone is shared, and adversarial discrimination and multi-class lesion classification are done simultaneously.The Generator generates realistic class specific dermoscopic images based on the seven lesion classes of the HAM10000 dataset while the Dual-Head Discriminator is used for adversarial discrimination and multi-class lesion classification using a shared feature extraction backbone. The proposed learning method is a combined method that enables the sharing of a common classification tree, but at the same time guarantees the different learning tasks the same visual representation that will be acquired during the learning process.

Limited number of annotated dermoscopic images and strong class imbalance between different skin lesion types were two important challenges addressed in the proposed framework for automatic skin cancer diagnosis. The model generates realistic synthetic images of all seven classes of the lesions, with the resulting diversity of the dataset complementing the diversity of images in the real dataset, and more examples may strengthen future deep learning classifiers. By maintaining the critical properties of lesions such as color, texture, shape, and pattern, the images can be used for data augmentation and research purposes.

Experimental results reveal that the proposed model is able to classify the multiple classes with good image synthesis quality. The quantitative analysis, based on the Accuracy, Precision, Recall, F1-Score, Balanced Accuracy Score (BAS), Approximate Fréchet Inception Distance (FID) and Inception Score, demonstrates the ability to learn meaningful representations for the image generation and the lesion classification tasks. In addition, the discriminator-based image quality evaluation offers an extra degree of confidence in the similarity of the generated image with the real image in dermoscopy. These results show that the shared learning strategy is able to effectively optimize the image generation of adversaries and diagnostic classification while maintaining a low computational complexity.

The other crucial aspect of this work is that the trained classification head of Dual-head Discriminator is directly utilized for real-time skin cancer detection. The proposed system does not need to train a standalone classifier, but rather utilizes the learnt discriminator features for the classification of the uploaded images into one of the seven classes of lesions and maps the images into a clinically meaningful Cancer-Positive or Benign prediction. This design can reduce the complexity of the model, accelerate the inference time of the model, and enhance the practicability of the model deployment. Applying the proposed system to a developed interactive web application further shows its applicability by permitting the user to produce synthetic images of skin lesions, upload dermoscopic images for diagnosis and have the prediction results displayed to them in real time via a simple web application.

The proposed framework also allows for efficient medical research, while maintaining privacy. It enables safe sharing of patient information and enables joint research among healthcare providers without revealing the patient's data.To minimize the concern about privacy while enabling the secure transfer of patient information for collaborative research in healthcare institutions. These synthetic datasets can also be leveraged to further improve future DL models, especially for rare categories of lesions where clinical data is limited.

The proposed scheme has good performance; however, there are some drawbacks. This current implementation results in dermoscopic images of 64×64 pixels, which may not be able to include fine-grained clinical details that would be of benefit to make an expert diagnosis. The experimental evaluation is based on one public dataset to assess the model's generalizability to other clinical datasets that are larger and more clinically diverse. Furthermore, although the created images have realistic visual features, more clinical studies by expert dermatologists are needed before using these synthetic images for routine clinical use.

The proposed framework will be further developed in the future to improve the image generation quality and diagnostic capabilities. Higher-resolution conditional GAN architectures and other recent generative models, such as StyleGAN and Diffusion Models will be used to gain more detailed and clinically realistic dermoscopic images. Various explainable AI (xAI) methods will be incorporated to improve interpretability of classification decisions as well as increase clinician trust, including Grad-CAM and attention visualization techniques. The multimodal learning approach, using dermoscopic image and patient data (age, sex, site, clinical history), will also be evaluated as future work in order to further enhance the diagnostic accuracy. Moreover, federated learning coupled with GAN-based anonymization method will be explored for cross-institutional collaborative training while maintaining patient privacy. Additionally, the proposed framework will be clinically evaluated by dermatologists, and will be included in cloud and mobile healthcare platforms to make the framework more practical to be implemented in real world skin cancer screening applications.

In summary, the proposed Conditional DCGAN framework demonstrates the feasibility and efficiency of the combined framework of conditional image synthesis and skin cancer classification. The framework offers a promising avenue towards the development of robust, scalable and privacy preserving computer-aided diagnostic systems to assist dermatologists in early diagnosis of skin cancer and to produce high quality synthetic medical images for future research and data augmentation.

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